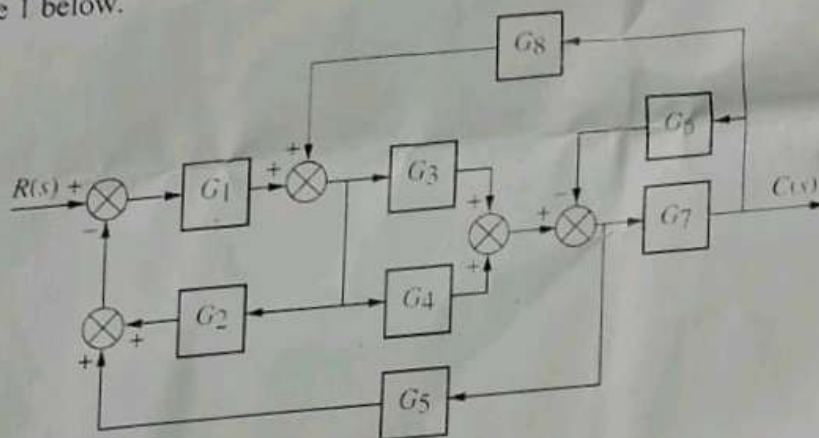
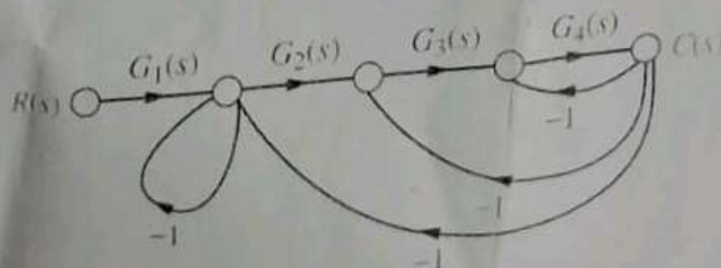


- (b) Obtain the transfer function $\frac{C(s)}{R(s)}$ for the system shown in the block diagram given in figure 1 below.



- shown in figure below.



- (b) Check the stability of a system with characteristic polynomial $p(s) = s^4 + 4s^3 + s^2 - 6s$ using Routh-Hurwitz array method. 3
- (c) Explain the working of a stepper motor in details using neat sketches. 3
4. (a) Find the range of gain K for a unity feedback system $G(s) = \frac{K}{s(s+2)(s+4)}$ that will make it stable, unstable and marginally stable using root locus. Assume $K > 0$ 5
- (b) What do you understand by phase-lead compensation. Discuss its effects and limitations briefly. Discuss Phase lead compensation using either Bode plots or Root locus diagram. 5
5. (a) Explain the Cauchy's argument principle 3
- (b) Draw the Nyquist plot and assess the stability of the closed loop system having open loop transfer function $G(s)H(s) = \frac{10}{s(s+1)}$. Also compute the phase and gain margin. 7
6. (a) Derive the expression for the unit step response of a system with $G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$. Also obtain the expression for peak overshoot and settling time for 5% tolerance band. 6
- (b) A unity feedback system with open loop transfer function $G(s) = \frac{10}{s(s+4)}$ is driven by a unit step input, compute suitable steady state error constant as well as steady state error due to this input. 4
7. (a) Sketch the Bode plot for a system with open loop transfer function $G(s) = \frac{10(s+2)}{s(s+4)}$ and compute the phase and gain margin. 5
- (b) A system with open loop transfer function $G(s) = \frac{4}{s(s+2)}$ is operating under unity feedback configuration. Find the value of peak overshoot, settling time for 2% tolerance band when excited with a unit step signal. 5
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